

ADVANCED SQL-RANKING AND WINDOW FUNCTION

Exercise 1: Ranking and Window Functions Goal: Use ROW_NUMBER(), RANK(), DENSE_RANK(), OVER(), and PARTITION BY. Scenario: Find the top 3 most expensive products in each category using different ranking functions. Steps: 1. Use ROW_NUMBER() to assign a unique rank within each category. 2. Use RANK() and DENSE_RANK() to compare how ties are handled. 3. Use PARTITION BY Category and ORDER BY Price DESC.

Step 1: Create Product Table

```
CREATE TABLE Products (  
    ProductID INT PRIMARY KEY,  
    ProductName VARCHAR(100),  
    Category VARCHAR(50),  
    Price DECIMAL(10,2)  
);
```

Step 2: Insert Sample Data

```
INSERT INTO Products VALUES  
(1,'Laptop A','Electronics',80000),  
(2,'Laptop B','Electronics',75000),  
(3,'Laptop C','Electronics',75000),  
(4,'Phone A','Electronics',50000),  
(5,'Shirt A','Fashion',2500),  
(6,'Shirt B','Fashion',2000),  
(7,'Shirt C','Fashion',2000),  
(8,'Shoes A','Fashion',4000),  
(9,'Table A','Furniture',10000),  
(10,'Chair A','Furniture',5000),  
(11,'Sofa A','Furniture',25000),  
(12,'Bed A','Furniture',25000)
```

Query 1: ROW_NUMBER()

```
SELECT *  
FROM (  
    SELECT ProductID,  
           ProductName,  
           Category,
```

```
Price,  
ROW_NUMBER() OVER  
(  
    PARTITION BY Category  
    ORDER BY Price DESC  
) AS RowNum  
FROM Products  
) RankedProducts  
WHERE RowNum <= 3;
```

Output Concept

Electronics

```
1 Laptop A 80000 Rank 1  
2 Laptop B 75000 Rank 2  
3 Laptop C 75000 Rank 3
```

Query 2: RANK()

```
SELECT *  
FROM (  
    SELECT ProductID,  
           ProductName,  
           Category,  
           Price,  
           RANK() OVER  
           (  
               PARTITION BY Category  
               ORDER BY Price DESC  
           ) AS RankNo  
    FROM Products  
) RankedProducts  
WHERE RankNo <= 3;
```

Example

```
80000 → Rank 1  
75000 → Rank 2  
75000 → Rank 2  
50000 → Rank 4
```

Query 3: DENSE_RANK()

```
SELECT *
FROM (
    SELECT ProductID,
           ProductName,
           Category,
           Price,
           DENSE_RANK() OVER
           (
               PARTITION BY Category
               ORDER BY Price DESC
           ) AS DenseRankNo
    FROM Products
) RankedProducts
WHERE DenseRankNo <= 3;
```

Example

80000 → Rank 1

75000 → Rank 2

75000 → Rank 2

50000 → Rank 3

ROW_NUMBER()

- Assigns a unique sequential number.
- Does not handle ties specially.

Example:

100

100

90

Ranks:

1

2

3

RANK()

- Same values get the same rank.
- Next rank is skipped.

Example:

100
100
90

Ranks:

1
1
3

DENSE_RANK()

- Same values get the same rank.
- No rank gaps.

Example:

100
100
90

Ranks:

1
1
2